Bio-Inspired Celestial Compass for Aircraft Navigation

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Outline

Motivation

Currently there are only two **self-sufficient** means for computation of bearing that has been widely accepted in technical systems [1]:

- **Magnetic compass**:
  - Suffers from distortion of the magnetic field by electric currents in the airframe
  - Geomagnetic field is not uniform which requires additional compensation scheme
- **Gyrocompass**:
  - Have strong anti-interference ability but accumulate error over time [2]
  - Relative position according to the initial position instead of absolute position
  - Long time to stabilize in case of a power failure

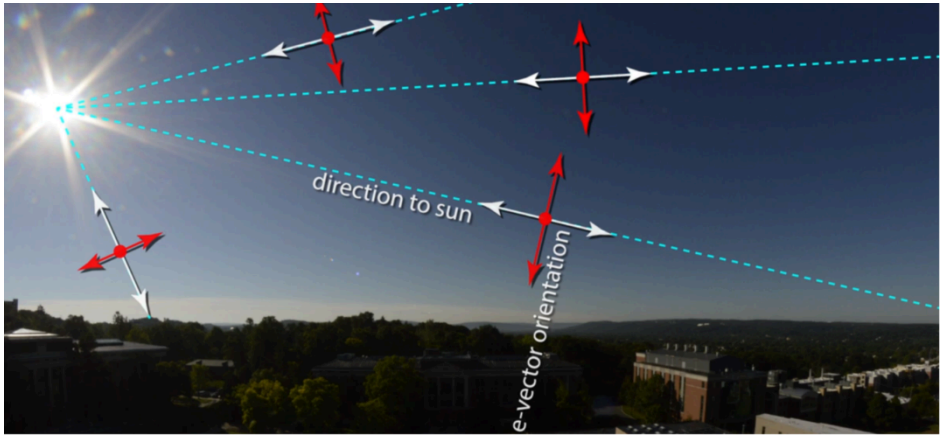
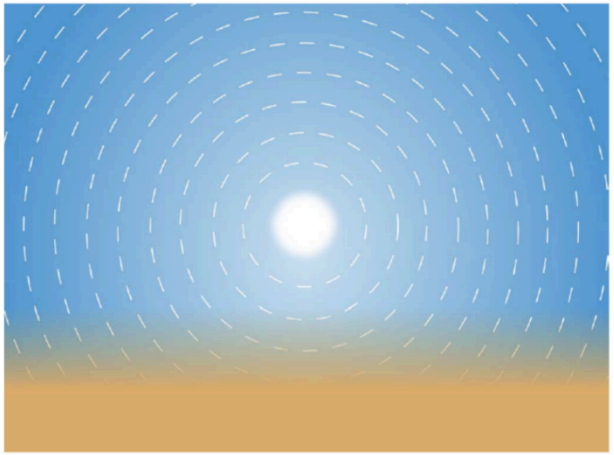
[1] A. Dawadee, Towards an optical aircraft navigation system, Applied Mechanics and Materials 629: 321-326 (2014).
[2] J. Liu et al, Global positioning method based on polarized light compass system, Review of scientific instruments 89: 054503 (2018).

A solution inspired by nature...

Case study: The Sahara desert ant

- Desert environment poses existential challenges:
  - Pheromone trails employed by other ant species for navigation cannot survive high temperature
  - No visual cues for use as path markers
  - Foraging durations are limited by intense heat
- Despite challenges, the ant manages to thrive!
- The remarkable navigation ability is achieved with the limited resources available

What is the underlying physical principle?

Polarization of Skylight

- When sunlight passes through the atmosphere, it becomes partially polarized
- Electric field for the polarized light becomes tangent to the sun's circumference
- Polarization becomes stronger with increasing distance from the sun
- The position of sun can be determined from analyzing the polarization of light